

SPARK Sports Academy Practice SBA

Full School-Based Assessment project brief

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: SPARK Sports Academy

Sector: Sports

Description of Project

SPARK Sports Academy offers after-school training in athletics, football, swimming and netball. The academy wants one system to manage athletes, guardians, programmes, monthly fees and training attendance. Management also wants a clear summary of income and programme participation before the new term begins.

Build a connected solution that stores member records, analyses fees and participation, communicates with guardians, presents the academy to families and automates a membership-fee calculation.

How to use this practice SBA

- Read the whole project before starting any software task.
- Complete the sections in order so information remains consistent from one file to the next.
- The wording and data are original SPARK practice material. Your teacher's current assignment takes priority.
- The practice task headings follow the style of common CSEC school assignments. Current CXC limits should still be followed.

Contents

Section	Practice weighting
Database Management	20 practice marks
Spreadsheet	20 practice marks
Word Processing	20 practice marks
Web Page Design	10 practice marks
Problem-Solving and Programming	15 practice marks

Database Management - 20 practice marks

Task A - Tables

1. Create a database named SportsAcademy_YourFullName.accdb.
2. Create the Athlete, Guardian and Programme tables using suitable field names, data types and primary keys.
3. Enter realistic records and apply the validation rules required by the project.
4. Create the required relationships and enforce the correct links between the tables.

Task B - Queries and Form

1. Create a query called Junior Athletes. Show AthleteID, athlete name, age and programme for athletes under 13. Sort by LastName ascending.
2. Create a query called Athlete Finance. Join Athlete and Programme. Show athlete name, programme, MonthlyFee and TermFee.
3. Create Athlete Information Form with Guardian subform and Previous/Next navigation buttons.

Task C - Report

1. Create Programme Fees Report grouped by ProgrammeName, sorted by LastName, with average TermFee for each programme.

Spreadsheet - 20 practice marks

Task A - Calculations and Functions

1. Create AcademyFees_YourFullName.xlsx with worksheets named Members and Summary. Enter or import the project data into Members and format it as a clear data table.
2. Create the main calculations using efficient formulas. The SPARK reference uses formulas such as `=D2*3`, `=IF(E2="Junior",F2*10%,0)` and `=F2-G2+$M$2`.
3. Use at least three suitable functions. Functions used in the SPARK reference include SUM, AVERAGE, IF, COUNTIF, VLOOKUP.

Task B - Analysis and Presentation

1. Sort and filter the data to answer this requirement: Show Active athletes in Swimming or Football only.
2. Create a summary showing Pivot-style summary of outstanding Balance by Programme. Then create a suitable chart.
3. Create a linked total or key result on the Summary worksheet so that it updates when the detailed sheet changes.

Word Processing - 20 practice marks

Task A - Fillable Form

1. Create a document named Athlete Registration Form_YourFullName.docx. The form must collect the information needed for this project and include clear instructions.
2. Use suitable form controls. The SPARK reference uses: Athlete name text box; DOB date picker; Programme drop-down list; Medical-information check box.

Task B - Mail Merge

1. Create a professional main document called Guardian Fee Notice. Use the project data as the mail-merge source.
2. Insert the following merge fields where they make sense: GuardianName, AthleteName, ProgrammeName, TermFee, Balance.
3. Complete the final merged output and include one relevant table, chart or other imported item from the connected SBA work where appropriate.

Web Page Design - 10 practice marks

Web Page Design

1. Create one web page for SPARK Sports Academy. The page must be suitable for the intended audience and include a project logo.
2. Organise the page into these content sections: About the Academy, Programmes and Fees, Training Schedule, Registration, Contact.
3. Add at least two suitable hyperlink types and test them.
4. Add at least one relevant graphic and make sure the page remains readable at different screen sizes.
5. Save, test and prepare the final web-page evidence.

Problem-Solving and Programming - 15 practice marks

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the amount each athlete should pay for a term and the total expected income for all athletes entered in one program run.
2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as AcademyFees_YourFullName.pas.
2. Test the program and capture evidence that the expected and actual results agree.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

Submission checklist

- SportsAcademy_YourFullName.accdb
- AcademyFees_YourFullName.xlsx
- Athlete Registration Form_YourFullName.docx
- Guardian Fee Notice_YourFullName.docx
- One completed web page with its supporting files
- AcademyFees_YourFullName.pas
- ProgramDocumentation_YourFullName.pdf

Final reminder

Keep the same names, codes, dates, fees and rules across your database, spreadsheet, word-processing, web and programming work. A strong SBA reads like one connected project.